

Carbon Chemistry

Friday, 31 July 2020 9:22 AM

Important Terms

- Saturated: All C-C bonds single, maximum possible number of atoms bonded to C atoms
- Hydrocarbon: Molecule containing only carbon and hydrogen atoms covalently bonded
- Homologous Series: A group of compounds with the same general formula, functional group, similar chemical properties, and members that differ from each other by CH_2
- Functional Group: A reactive part of a molecule which largely determines its chemical properties (reactions)

Alkanes

- Homologous series with general formula $\text{C}_n\text{H}_{2n+2}$
- Saturated hydrocarbons

Physical Properties

- $\text{C}_1\text{-C}_4$: Gas
- $\text{C}_5\text{-C}_{20}$: Liquid
- C_{21+} : Solid

Chemical Properties

- C-C bonds are extremely strong
- C-H bonds are strong, although not as strong as C-C bonds
- Alkanes are chemically inert (don't react without specific conditions)

Reactions

Combustion

- Combustion: A reaction with oxygen
- Complete Combustion (Excess oxygen)
 - $\text{Alkane} + \text{O}_{2(g)} \rightarrow \text{H}_2\text{O}_{(g)} + \text{CO}_{2(g)}$
- Incomplete Combustion (Limited oxygen)
 - $\text{Alkane} + \text{O}_{2(g)} \rightarrow \text{H}_2\text{O}_{(g)} + \text{CO}_{(g)}$
 - May also produce soot: $\text{Alkane} + \text{O}_{2(g)} \rightarrow \text{H}_2\text{O}_{(g)} + \text{C}_{(s)}$
- Conditions: Spark
- Exothermic (releases energy)
 - Energy comes from bond formation

Substitution - Halogenation

- Reagents: Alkane + Halogen (Cl_2 , Br_2 , I_2)
 - As liquids/gases halogens are toxic, we use them in aqueous form ('Chlorine water', 'Bromine water', 'Iodine water')
- Conditions: UV light
- One hydrogen atom is replaced by a halogen
- Observation: E.g. $\text{C}_2\text{H}_{6(g)} + \text{Br}_{2(aq)} \rightarrow \text{C}_2\text{H}_5\text{Br} + \text{HBr}_{(aq)}$
 - Colourless gas bubbled through orange solution and exposed to UV light, solution turns colourless

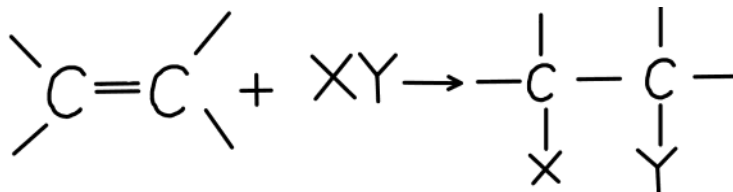
Alkenes

- Homologous series with general Formula: C_nH_{2n}
- $\text{C}=\text{C}$ double bonds
- The bonds in the double bond are not identical
 - Sigma bond (σ): Directly overlaps many orbitals, very strong, does not react easily
 - Pi bond (π): Indirectly overlaps orbitals, very weak, reacts easily
 - Doesn't need catalyst, light, etc.

Reactions

- **Addition reactions**

- General Formula:



- When two non-symmetrical reactants are added to each other, they will form isomeric products (products will have potential isomers)

Hydrogenation (aka Saturation)

- Addition with H_2
- Conditions:
 - Nickel catalyst
 - High temperature + pressure

Halogenation

- Addition with Cl_2 , Br_2 or I_2
- No conditions (can even take place in the dark)
- Reaction with $\text{Br}_{2(\text{aq})}$ is generally used to test for alkenes
 - E.g. Ethene + Bromine water $\rightarrow$ 1,2-dibromoethane
 - ◆ $\text{CH}_2\text{CH}_2 + \text{Br}_2 \rightarrow \text{CH}_2\text{BrCH}_2\text{Br}$
 - Observations: Colourless gas added to orange solution, solution turns colourless

Hydrogen Halides

- HCl , HBr , HI (NOT ACIDS)
- E.g. $\text{CH}_2\text{CH}_2 + \text{HI} \rightarrow \text{CH}_3\text{CH}_2\text{I}$
 - Ethene + Hydrogen Iodide $\rightarrow$ Iodoethane
- No specific conditions, can also react in the dark

Hydration (Water)

- Addition of water
- Conditions:
 - Catalyst (H_3PO_4)
 - High temperature and pressure
- $\text{CH}_2\text{CH}_2 + \text{H}_2\text{O} \rightarrow \text{CH}_3\text{CH}_2\text{OH}$
 - Ethene + Water $\rightarrow$ Ethanol

Alkynes

- Homologous series with formula: $\text{C}_n\text{H}_{2n-2}$
- $\text{C}\equiv\text{C}$ triple bond
- Very reactive as it has 2 π bonds
 - Add 2H_2 for saturation as it requires 4 H atoms to fully saturate

Bromine

- ONLY REACTS WITH PURE HALOGENS - doesn't react with e.g. Bromine water
- Needs a catalyst depending on the halogen it is reacting with

Nomenclature

1. If has geometric isomerism, 'cis-' or 'trans-'
2. Order halides and alkyl substituents alphabetically (e.g. Bromo, Ethyl, Methyl)
3. Add prefixes to the groups
 - a. Numbers to show which Carbon the group is attached to (e.g. 1-bromo)
 - b. 'di-', 'tri-', 'tetra-', etc. if multiple of one halide/substituent
 - E.g. 1,4-Dibromo, 2,3,3-trimethyl, etc.
4. Add the base name
 - a. 'Cyclo-' if it is cyclic
 - b. Prefix to show number of carbons (meth, eth, prop, but, etc.)
 - c. Number to show where alkene/alkyne group(s) are (if any)

- d. '-ane', '-ene', or '-yne' to show if it is an alkane, alkene or alkyne
- Examples
 - Benzene: cyclohex-1,3,5-ene
 - 1,2,4,5-tetramethyl-3-propylcyclopentane
 - 3-ethyl-2,4,6-trimethylnonane

Isomers

- Isomers all have the same molecular formula

Structural

- Different structural formula (different bonding between atoms)
- E.g. Position of functional groups, branching

Geometric

- Must have alkene group
 - Double bonds non-rotational, different bond orientation = different position of atoms in 3D space
- *Cis*-isomers = same side ; *Trans*-isomers = different sides

Combustion Test

- **Cyclohexane**: Colourless liquid set on fire, burns well producing bright orange flame, soot and smoke
- **Cyclohexene**: Colourless liquid set on fire, burns well producing bright orange flame, more soot/smoke than cyclohexane
- **Toluene (methylbenzene)**: Colourless liquid set on fire, burns well producing bright orange flame, and the most soot/smoke out of these 3
- Soot is unburned Carbon
- More soot is produced from the last two as they have a larger *Carbon : Hydrogen* ratio, so less carbon is burned and more is released as soot

Bromine Water Test

- **Alkanes**: Don't react chemically (need UV light) - "orange solution becomes paler, colourless liquid turns orange"
 - $\text{Br}_{2(\text{aq})}$ moves to the top layer (organic solvent) as they are both non-polar, so it dissolves better than in H_2O
- **Alkenes**: React when stirred, "Orange solution added to colourless liquid, orange solution decolourises forming two immiscible layers"
- **Toluene (methylbenzene)**: Doesn't react "orange solution remains orange, colourless liquid remains colourless"
 - Conclusion: No double bonds exist in benzene

Class

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Organic Molecules

- Saturated: All C-C bonds are single bonds; Maximum number of atoms possible are bonded
- Unsaturated: Not only C-C single bonds (double/triple)
- Hydrocarbon: Molecule that only contains hydrogen and carbon atoms

Alkanes

- Saturated hydrocarbons
- Part of a homologous series
 - Homologous Series:
 - Same general formula (C_nH_{2n+2})
 - One consecutive member differs from the previous by CH_2

Physical States

- Larger molecules have stronger dispersion forces
 - C1-C2: Gas
- C₃-C₄: Gas, easily liquified
- C₅-C₂₀: Liquid
- C₂₁+: Solid

Cycloalkanes

- Carbons make a 'ring' (no end)
- Not in the same homologous series as alkanes
 - General Formula: C_nH_{2n}

Reactions

- C-C bonds are extremely strong
- C-H bonds are strong, although not as strong as C-C bonds
- Alkanes are chemically inert (generally don't react)

Combustion

- Combustion: A reaction with oxygen
- Exothermic (releases energy)
 - Energy comes from bond formation
- As energy is required to break bonds, it has a high activation energy (spark)
- Complete Combustion (Excess oxygen)
 - $\text{Alkane} + O_{2(g)} \rightarrow H_2O_{(g)} + CO_{2(g)}$
- Incomplete Combustion (Limited oxygen)
 - $\text{Alkane} + O_{2(g)} \rightarrow H_2O_{(g)} + CO_{(g)}$
 - May also produce soot: $\text{Alkane} + O_{2(g)} \rightarrow H_2O_{(g)} + C_{(s)}$
 - CO is called the 'silent killer' (colourless, odourless)
 - When inhaled, haemoglobin combines with CO and is poisoned, and the person dies

Nitrogen

- Nitrogen gas has a triple bond, very hard to break
- A combustion engine (in a car) has perfect conditions to break the bonds - high pressure and temperature
 - ◆ N reacts with O forming NO, N₂O or NO₂
 - ◆ NO₂ is extremely dangerous

Substitution

Halogenation

- Homologous Series: Halogenoalkanes
- Reagents: Alkane + Halogen (Cl₂, Br₂, I₂)

- Conditions: UV light
- As liquids/gases they are very toxic, so we use them in aqueous form
 - ◆ Called 'Chlorine water', 'Bromine water', 'Iodine water'

Colours

- ◆ $\text{Cl}_{2(\text{aq})}$: Pale yellow solution
- ◆ $\text{Br}_{2(\text{aq})}$: Orange solution
- ◆ $\text{I}_{2(\text{aq})}$: Brown solution
- ◆ E.g. $\text{C}_2\text{H}_6(\text{g}) + \text{Br}_{2(\text{aq})} \rightarrow \text{C}_2\text{H}_5\text{Br} + \text{HBr}_{(\text{aq})}$
 - ◇ Orange solution turns colourless when added to an alkane

Functional Groups

- Functional Group: A reactive part of a molecule that largely determines the chemical properties (reactions) of the molecule

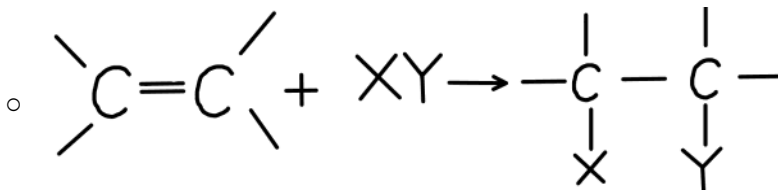
C=C	Alkene
NH ₂	Amine
OH	Hydroxyl
C(=O)OH	Carboxylic Acid

Alkenes

- General Formula: C_nH_{2n}
- C=C double bonds
 - Double bond is a *functional group*
- The double bonds are not identical
 - Sigma bond (σ): Directly overlaps many orbitals, **very strong, does not react easily**
 - **Pi bond (π): Indirectly overlaps orbitals, very weak, reacts easily**
 - Doesn't need catalyst, light, etc.

Chemical Properties

- All alkenes react in **addition reactions**
- General Formula



- When two non-symmetrical reactants are added to each other, they will form isomeric products (products will have potential isomers)

Hydrogenation (aka Saturation)

- Addition with H_2
- E.g. Ethene + $\text{H}_2 \rightarrow$ Ethane
- Needs catalyst/conditions (on top of arrow) to break H-H bonds:
 - Nickel catalyst
 - High temperature + pressure

Halogenation

- Addition with Cl_2 , Br_2 or I_2
- Reaction with $\text{Br}_{2(\text{aq})}$ is generally used to test for alkenes
 - E.g. Ethene + Bromine water $\rightarrow$ 1,2-dibromoethane
 - $\text{CH}_2\text{CH}_2 + \text{Br}_2 \rightarrow \text{CH}_2\text{BrCH}_2\text{Br}$
 - No conditions necessary (can even take place in the dark)
 - Observations: Colourless gas added to orange solution, solution **quickly?** turns colourless

Hydrogen Halides

- HCl, HBr, HI (NOT ACIDS)
- $\text{CH}_2\text{CH}_2 + \text{HI} \rightarrow \text{CH}_3\text{CH}_2\text{I}$

- Ethene + Hydrogen Iodide → Iodoethane
- No specific conditions

Hydration (Water)

- Addition of water
- High activation energy - conditions:
 - Catalyst (H_3PO_4)
 - High temperature and pressure
- $\text{CH}_2\text{CH}_2 + \text{H}_2\text{O} \rightarrow \text{CH}_3\text{CH}_2\text{OH}$
 - Ethene + Water → Ethanol
- Hydrocarbon Cracking: Breaking long-chain hydrocarbons into multiple short-chain hydrocarbons

Alkynes

- General formula: $\text{C}_n\text{H}_{2n-2}$
- $\text{C}\equiv\text{C}$ triple bond

Physical Properties

- Insoluble in water - will always form immiscible layers

Chemical Properties

- Very reactive
 - 2 π bonds which are both very reactive

Reactions

Saturation

- Propyne + Hydrogen → Propane
 - $\text{C}_3\text{H}_4 + 2\text{H}_2 \rightarrow \text{C}_3\text{H}_8$

Combustion Tests

- **Cyclohexane**: Colourless solution set on fire, burns well producing bright orange flame, soot and smoke
- **Cyclohexene**: Colourless solution set on fire, burns well producing bright orange flame, more soot/smoke than cyclohexane
- **Toluene (methylbenzene)**: Colourless solution set on fire, burns well producing bright orange flame, and the most soot/smoke out of these 3
- Soot is unburned Carbon
- More soot is produced from the last two as they have a larger *Carbon : Hydrogen* ratio, so less carbon is burned and more is released as soot

Bromine Water Test

- "Orange solution added to colourless liquid, forms two immiscible layers..."
- Alkanes: Don't react chemically (need UV light) - "orange solution becomes paler, colourless liquid turns orange"
 - $\text{Br}_{2(\text{aq})}$ moves to the top layer (organic solvent) as they are both non-polar, so it dissolves better than in H_2O
- Alkenes: React when stirred, decolourises solution "orange solution turns colourless"
- Toluene (methylbenzene): Doesn't react "orange solution remains orange, colourless liquid remains colourless"
 - No double bonds exist in benzene

Isomers

- Isomers all have the same molecular formula

Structural: Different structural formula

- Branching
- Positions of functional group

Geometric: Different orientation of double bonds and different arrangement of atoms in space (only for alkenes)

- 'Cis-' = same side; 'Trans-' = opposite side
- Must have C=C double bond (double bonds are non-rotational)
- Different groups attached to each Carbon atom on the C=C double bond
- Naturally all isomers will be cis isomers

Ways of Drawing (only allowed to use structural)

Displayed Formula: Show all bonds

Semi-condensed Formula: Displayed, but join CH_n together

Condensed Formula:

- E.g. Octane: CH₃(CH₂)₆CH₃
- Sub-chains MUST be in brackets

Skeletal: Lines represent carbons and hydrogens

Benzene

- Part of a homologous series
 - Benzene: C₆H₆
 - Methylbenzene: C₇H₈

Physical Properties

- Liquid at room temp (relatively large molecule)
- Insoluble in water

Reactions

- **Combustion:** $2\text{C}_6\text{H}_6 + 15\text{O}_2 \rightarrow 12\text{CO}_2 + 6\text{H}_2\text{O}$
- **Substitution:** $\text{C}_6\text{H}_6 + \text{Br}_2 \rightarrow \text{C}_6\text{H}_5\text{Br} + \text{HBr}$
 - **Conditions:** (not on syllabus)
 - Must be *pure* bromine (not bromine water)
 - FeBr₃ catalyst (for bromine) - different for Cl and I

12.1 - Hydrocarbon Sources and Uses

- Hydrocarbon: Compounds containing only Hydrogen and Carbon
- Fractional Distillation used to separate mixtures of hydrocarbons into fractions of similar temperatures
- They can be used as fuels, lubricants, asphalt and solvents, or converted into other materials such as plastics, synthetic fibres and pharmaceuticals

12.3 - Alkanes: Structure and Nomenclature

- General Formula: C_nH_{2n+2}
- *Saturated* hydrocarbons
- Suffix '-ane'
- May be straight chain or branched

12.4 - Alkanes: Structural Isomerism

- Structural Isomers: Compounds with the same molecular formula but a different structural formula
- May have similar physical/chemical properties but are not identical
- E.g. $CH_3(CH_2)_3CH_3$ vs $CH_3CH_2CH(CH_3)CH_3$ vs $CH_3C(CH_3)_2CH_3$

12.5 - Alkenes: Structure and Nomenclature

- General Formula: C_nH_{2n}
- Unsaturated hydrocarbons with at least one Carbon-Carbon double bond
- Suffix '-ene'

12.6 - Alkenes: Geometric Isomerism

- Double bonds cannot rotate, so even with the same structural formula (position of bonds), there may exist molecules with different orientation of atoms
- *Cis*-isomers have both Carbon atoms on the both side
- *Trans*-isomers have both Carbon atoms on different sides

12.7 - Cycloalkanes

- General Formula: C_nH_{2n}
- 3+ Carbon atoms arranged into a close ring structure
- Prefix 'cyclo-' added to stem name
- E.g. Cyclopropane, Cycloheptane

12.8 - Benzene

- C_6H_6
- Stable, flat, hexagonal cyclic structure
- All Carbon-Carbon bonds are identical in nature
 - Length between that of a single and double bond
- Structure is incorporated into molecules of many natural substances
 - These are called *aromatic* as they have a distinct odour
- Occurs naturally in coal and crude oil, also produced by burning natural materials (e.g. Forest fires)
- Used in plastics, detergents, pesticides and petrol
- Use limited as some of its aromas are known to be cancer-causing

12.9 - IUPAC Nomenclature for Alkanes and Alkenes

1. Find the longest continuous chain of Carbon atoms. This is the parent chain and determines the stem name
 - If a C=C double bond is present, the parent chain **must** include it
2. Number the Carbon atoms sequentially so that the substituent/halogen group has the lowest possible number
 - If it is an alkene, the double bond has priority over alkyl groups/halogen groups
3. Prefixes ordered alphabetically and numbered according to which Carbon atom they are bonded to
 - Prefixes ('di', 'tri', 'tetra') are used for each group if multiple groups are present

Miscellaneous

- Numbers separated by commas, numbers and letters separated by a hyphen
- Alkyl substituents are named with the prefixes 'methyl', 'ethyl', 'propyl', etc.
- Halogen groups are named with the prefixes 'fluoro-', 'chloro-', 'bromo-', 'iodo-'

12.10 - Addition Reactions of Hydrocarbons

- As alkenes are unsaturated (bonding capacity not fully utilised), they readily undergo addition reactions with the double C=C bond
 - The double bond is replaced with other atoms such as H, F, Cl, Br or I
- Alkanes and benzene do NOT undergo addition reactions
- Reagents for addition reactions include: H₂, Cl₂, Br₂, HCl, HBr, and HI
- E.g. $\text{CH}_2=\text{CH}_2 + \text{HCl} \rightarrow \text{CH}_3\text{CH}_2\text{Cl}$
- If adding an acid, the added H atom will generally attach to the carbon of the double bond which already has the most bonded H atoms (Markovnikov's rule)

12.11 - Substitution Reactions

- Occurs when an alkane/benzene reacts with Cl₂ or Br₂
- One+ H atoms are replaced by the halogen
- Much slower than addition reactions - can be used to distinguish between saturated/unsaturated hydrocarbons
- If excess halogens are used, complete substitution can occur (e.g. Forming CCl₄ from CH₄)

12.12 - Combustion of Hydrocarbons

- Complete combustion: $\text{Hydrocarbon} + \text{O}_2 \rightarrow \text{H}_2\text{O} + \text{CO}_2$
- Incomplete combustion: $\text{Hydrocarbon} + \text{O}_2 \rightarrow \text{H}_2\text{O} + \text{CO}$
 - Bad because:
 - Inefficient as it produces significantly less energy than complete combustion
 - Health hazard due to toxic products formed (CO)
 - Attaches to haemoglobin molecules in blood, depriving cells of oxygen
 - Colourless and odourless, 'silent killer'

13.8 - Fuels: The Energy Source for a Modern Economy

- Fuel: Substances that react with oxygen to produce heat
- 2 types of fuels used today:
 - Fossil fuels (e.g. Coal, oil, natural gas)
 - Biofuels (e.g. Biodiesel, bioethanol, biogas)
- Different fuels produce different amounts of energy and CO₂ emissions

13.9 - Comparing Fossil Fuels

- Two aspects of fuel are its fuel value and carbon emissions
 - Fuel value: Compare energy released from complete combustion of equal masses/volumes of different fuels
 - E.g. kJ g⁻¹, MJ kg⁻¹, MJ L⁻¹
 - For pure fuels it can be calculated from the standard heat of combustion (ΔH_c) of

- the fuel
- Most fuels are mixtures, so the fuel value must be determined experimentally
- Fuel Value = $\Delta H/M$
 - ΔH = standard heat of combustion of 1 mole
 - M = mass burnt (molar mass as 1 mole being burnt)
- Carbon Emissions: Compares amount of CO_2 produced when a given amount of energy is released
 - $\text{g}(\text{CO}_2) \text{ MJ}^{-1}$; Grams CO_2 per Megajoule
 - Coal is the cheapest and most abundant fossil fuel, but has the highest carbon emissions
 - Global move to replace coal with natural gas
 - However, natural gas has a lower fuel value

13.10 - Biofuels: A Low Emission Alternative

- Produced from biomass, renewable
- Lower CO_2 emissions than fossil fuels
- E.g. Ethanol ($\text{C}_2\text{H}_5\text{OH}$) can be produced by fermentation of plant sugars (yeast catalyst)
 - $\text{C}_6\text{H}_{12}\text{O}_6 \rightarrow 2\text{C}_2\text{H}_5\text{OH} + 2\text{CO}_2$
 - As glucose is a product of photosynthesis, so all of the carbon originates from atmospheric CO_2
 - Not completely 'carbon neutral' as CO_2 is still emitted from fossil fuels to make biofuels

8.1 - Alkanes

- Carbon can form many compounds because:
 - 4 valence electrons (potential 4 atoms bonded)
 - Can form strong covalent bonds to other C atoms
 - Bonds can form as single, double, or triple bonds
- Many of these compounds are produced from crude oil
 - Crude oil produced from pressure on decayed organic matter over millions of years
- Cracking
 - Large hydrocarbon molecules broken into smaller molecules using heat and a catalyst
- Homologous series: Each member differs by CH_2 , similar chemical properties, general formula, functional group

8.2 - Alkenes

- At least one $\text{C}=\text{C}$ double bond
- C_nH_{2n}
- Simplest one is ethene (C_2H_4) as there must be at least two C atoms
- Double bonds cannot rotate freely
 - Cis isomers have the same group on the same side of the double bond
 - Trans isomers have the same group on opposite sides of the double bond
- When naming, the longest chain must include the double bond and carbons are numbered starting from the end closest to the double bond

8.3 - Benzene

- C_6H_6 ring with delocalised electrons that can move throughout the ring
- Carbon-carbon bonds are all the same length
- Molecules containing benzene are known as 'aromatic', as they usually have a distinct odour
 - 'Arenes' used for aromatic hydrocarbons

8.4 - Reactions

- Alkane + Halogen: UV light
- Benzene + Halogen/Alkyl Halide: Catalyst
- Alkene + H_2 : Ni catalyst
- Alkene + Halogen: None (even in the dark)
- Alkene + Hydrogen Halide: None
- Alkene + Water: H_3PO_4 catalyst + High temp/pressure
- Bromine used to test for unsaturation

Combustion

- Complete: Plenty of oxygen, products are $\text{CO}_2 + \text{H}_2\text{O}$
- Incomplete: Limited oxygen, products are $\text{CO} + \text{H}_2\text{O}$
 - CO is the 'silent killer'

Class

Thursday, 20 August 2020 12:05 PM

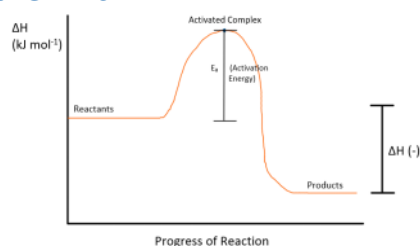
Introduction

- To react, reactant bonds must break and product bonds must form
 - Breaking bonds is endothermic (takes energy from surroundings)
 - Forming bonds is exothermic (releases energy into surroundings)
- **System:** Chemicals taking part in the reaction and their bonds
 - A, B and AB
- **Surroundings:** The whole universe surrounding it
 - E.g. In a solution: Most to least immediate - water, beaker, air, etc.
- **Activation Energy:** Minimum amount of energy required for reaction to occur

Exothermic vs Endothermic

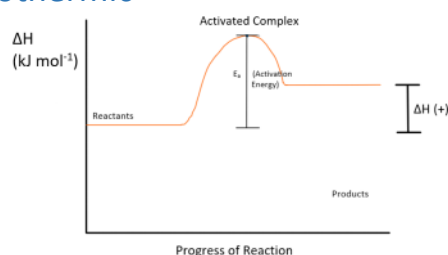
Feature	Endothermic	Exothermic
ΔH	Positive	Negative
Product Bond Strength (compared to reactants)	Stronger	Weaker
Surrounding Temperature	Decreases	Increases
Heat/Energy	Reactants	Products

Exothermic



- E.g.¹ $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O} + \text{Heat}$
- E.g.² $\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3$
 - Very high activation energy as nitrogen has triple bonds

Endothermic



- E.g.¹ $\text{CaCO}_3 + \text{Heat} \rightarrow \text{CaO} + \text{CO}_2$

Catalysts

- Catalyst: A chemical which increases the rate of reaction by providing an alternative pathway with a lower activation energy
- DOES NOT LOWER ACTIVATION ENERGY OF REACTANTS - it is a fixed value

Enthalpy

- **Enthalpy (H):** Heat energy at constant pressure
 - Enthalpy cannot be measured - we measure energy change to calculate enthalpy
- ΔH : Change in enthalpy

- $\Delta H = H_{(\text{products})} - H_{(\text{reactants})}$

Enthalpy Change of Combustion (ΔH_c)

- **Enthalpy Change of Combustion (ΔH_c):** Enthalpy change when 1 mole of a compound reacts completely with oxygen (standard states and conditions)
 - Standard Conditions: 25°C & 1atm OR 298°K & 100kPa
- $\Delta H(J) = mc\Delta T$
 - m = mass of water heated (g)
 - c = specific heat capacity of water (4.18J g⁻¹ K⁻¹ OR 4.18J g⁻¹ °C⁻¹)
 - ΔT = temperature change in water (°K / °C)
- $\Delta H_c = \frac{\Delta H}{n(\text{fuel})}$
- Higher molar mass fuels are more efficient (higher ΔH_c)

Experiment to Calculate ΔH_c

- Metal tin of water

Problems

- Not all heat energy goes towards heating up the water
- Not complete combustion (soot formed)
- Delay between taking cap off and starting flame where fuel is evaporating

Improvements

- Insulating shield
- Constant air flow so complete combustion is more likely

Biofuels

- Biofuel: A fuel produced from biomass rather than other resources, such as fossil fuels
 - Renewable
 - Carbon-neutral

Carbon Neutral

- No net carbon footprint
- Carbon dioxide released by fuels into the atmosphere are taken from the atmosphere by the plants in the first place
- However, other fuels are required to create biofuels, so it is not completely carbon-neutral
 - Growing plants, transportation, manufacturing

Bioethanol

- Made from various plant materials
- Used as a blending agent with gasoline to increase octane and cut down carbon monoxide emissions

Biodiesel